

- There are two types of electric current (i) Alternating current (AC) and (ii) Direct current (DC).
- Alternating current is used in houses and factories and its frequency is 50 Hz.
- Ordinary DC ammeter and DC voltmeter cannot measure alternating current/voltages. They record zero reading, when used in AC circuits, because average value of alternating current/voltage over a full cycle is zero.

Galvanometer

It is a device used to detect and measure electric current in a circuit. It can measure current up to 10^{-6} A.

A galvanometer can be converted into a voltmeter by connecting a very high resistance in its series.

- **Note** Conventionally the direction of electric current is opposite to that of electrons. Since, electrons flow from negative terminal of a cell to its positive terminal, electric current flows from positive terminal to negative terminal.

Capacitor

A capacitor or condenser is a device over which a large amount of charge can be stored.

- Capacitance of capacitor (C) = $\frac{\text{Charge (Q)}}{\text{Potential (V)}}$
- Its unit is coulomb/volt or farad.
- A capacitor is used in several electrical devices having an electric motor and in several electronic circuits.

OHM'S LAW

If the physical circumstances of the conductor. (length, temperature, etc) remains constant then the current flowing through the conductor is directly proportional to the potential difference across its ends.

$$\text{i.e. } I \propto V$$

$$\text{or } V = IR, \text{ where } R \text{ is resistance.}$$

Resistance

The resistance of a conductor is directly proportional to its length and inversely proportional to its cross-sectional area. If l and A are respectively length and cross-sectional area of a conductor and R is its resistance, then

$$R \propto \frac{l}{A} \Rightarrow R = \rho \frac{l}{A}$$

Unit of resistance is ohm (Ω).

where, ρ is a constant of material of conductor is called specific resistance or resistivity. Its SI unit is ohm-metre.

- On increasing the temperature of the metal, its resistance increases.
- Those materials whose electrical conductivity lies in between that of insulators and conductors are called semiconductors. On increasing the temperature of semiconductor, its resistance decreases.

- The reciprocal of resistance of a conductor is called the electrical conductance of the conductor.

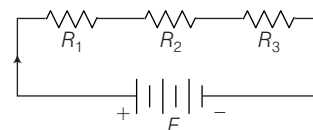
$$\text{Conductance} = \frac{1}{\text{Resistance}}$$

- Unit of conductance is mho or siemen.
- The specific resistance of the material depends only on the material of conductor and its temperature.
- Resistivity increases with temperature.
- Resistivity of a conductor change with impurity.
- Resistivity of an alloy is greater than the resistivity of its constituents and remains unchanged with temperature.
- If a wire is stretched or doubled on itself, its resistance will change, but its specific resistance will remains unaffected.
- The reciprocal of resistivity of a conductor is called its conductivity. Its SI unit is mho m^{-1} or siemen/metre ($S m^{-1}$).

Combination of Resistances

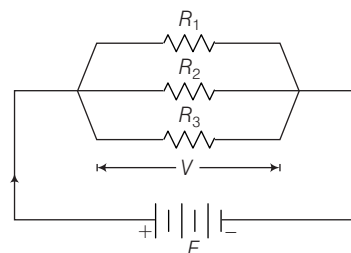
There are two combinations given below:

- (i) **Series combination** $R = R_1 + R_2 + R_3$ and here current flows through each conductor is same.



The total resistance in the series combination is more than the greatest resistance in the circuit.

- (ii) **Parallel combination** $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$ and here potential across each conductor is same.



The total resistance in parallel combination is less than the least resistance of the circuit.

ELECTRIC POWER

Electrical power is the electrical work done per unit time.

$$P = \frac{W}{t} \quad \text{or} \quad P = V \times I$$

$$\text{or} \quad P = I^2 R \quad [\because V = IR]$$

$$\text{or} \quad P = \frac{V^2}{R}$$

Its SI unit is watt (W).

- 1 kilowatt hour = 3600000 J = 3.6×10^6 J
- 1 Horse power (HP) = 746 W
- 1 Horse power (HP) = 550 foot-pound/second
- Unit = $\frac{\text{volt} \times \text{ampere} \times \text{hour}}{1000}$
- Unit = $\frac{\text{watt} \times \text{hour}}{1000}$

Heating Effect of Electric Current

Heat is produced, when electric current is passed through a conductor. This effect is called heating effect of electric current.

$$\text{Heat produced} = I^2 R t = V I t = \frac{V^2 t}{R} \text{ joule}$$

- **Note** The filament of an electric bulb gets very hot and therefore easily oxidised. Thus, bulbs are sealed with inactive nitrogen and argon gases to prevent oxidation of the filament and increase its life.
- In electric bulb, tungsten is used for the construction of filament because it has high melting point.

IMPORTANT POINTS BASED ON HEATING EFFECT

- In home appliances like electric iron, electric heater and heating rod, the heating element used is of a nichrome (an alloy of Ni and Cr) wire. Nichrome has high melting point and high resistivity. To avoid the risk of electric shock, the metal body of electrical appliances is earthed.
- An electric fuse is generally prepared from tin-lead alloy (63% tin + 37% lead). It should have high resistance and low melting point. It is connected in the series.
- Tubelight contains a long tube of glass which is linked internally with a fluorescent substance. It is filled with an inert gas like argon along with some mercury.

Magnetic Effect of Current

When electricity passes through insulated copper wire, it develops a magnetic field around it. Larger the amount of electricity, stronger the magnetic field. And when we place compass needle in the radius of the magnetic field it diverts from its North-South position.

MAGNETS

Magnet is a piece of iron or other materials that can attract iron containing objects and the property of attracting the magnetic substance by a magnet is called magnetism.

- The magnets which do not lose their magnetism with normal treatment are called permanent magnets.
- The materials which retain their magnetism for a long time are called hard magnetic materials.
- In bar, rod and horse-shoe magnets, North or South poles are either indicated by the letter *N* or *S*.

- When poles of two magnets are brought close together, they exert force on each other. This force is called intersection between the poles.
- If we cut a magnet in two parts, then each separate part will behave as a magnet.
- Permanent magnets are usually made of alloys such as carbon steel, chromium steel, cobalt steel, tungsten steel and alnico (alloy of aluminium, nickel, cobalt and iron).
- Such permanent magnets are used in microphones, loudspeakers, electric clocks, ammeters, voltmeters, speedometer and many other devices.
- **Note** The Magnetic Resonance Imaging (MRI) technique which is used to scan inner human body parts in hospitals also uses magnets for its working.

Magnetic Field

The space in the surrounding of a magnet or a current carrying conductor in which its magnetic effect can be experience is called magnetic field.

- Magnetic lines of force are imaginary lines in the magnetic field, which shows the direction of magnetic field continuously.
- The magnetic field lines originate from the North pole of a magnet and end at its South pole.
- The magnetic field lines do not intersect one another.

Magnetic Materials

According to behaviour of magnetic substances, they are classified into three classes

- Those substances when placed in an external magnetic field, acquire a very low magnetism in direction opposite to the field are called **diamagnetic substances**. e.g. copper, silver, bismuth, zinc, diamond, salt, water, mercury, etc.
- The permeability of diamagnetic substance is less than one.
- Those substances when placed in an external field, acquire a feeble magnetism in the direction of field are called **paramagnetic substances**. e.g. aluminium, sodium, potassium, oxygen, etc.
- The permeability of paramagnetic substance is slightly greater than one.
- Those substances, when placed in a magnetic field, acquire a strong magnetism in the direction of field are called **ferromagnetic substances**. These substances when brought near the end of a strong magnet get radially attracted towards it. e.g. iron, nickel, cobalt, magnetite, etc.
- The permeability of ferromagnetic substance is much greater than one.

Electromagnet

An electromagnet is a solenoid coil that attains magnetism due to flow of current. It works on the principle of magnetic effect of current. It is used in electric bells, electric motors, telephone, diaphragms, loudspeakers etc.

Magnetic Flux

The magnetic flux linked with a surface is equal to the total number of magnetic lines of force passing through that surface normally. Its unit is weber.

ELECTROMAGNETIC INDUCTION

When a change occurs in the magnetic flux linked with the coil, an emf is induced in the coil. The phenomenon is called electromagnetic induction.

- The phenomenon of production of induced emf in a circuit due to change in magnetic flux in its neighbouring circuit is called **mutual induction**.
- When the electric current flowing through a circuit changes, the magnetic field linked with circuit also change. As a result an induced emf is set up in the circuit. This phenomenon is called **self-induction**.
- Microphone converts sound energy into electrical energy and works on the principle of electromagnetic induction.

Electric Generator

It is a device which converts mechanical energy into electrical energy. An electric generator is based on the principle of electromagnetic induction. It is used to produce electric current by varying magnetic field through a coil.

- A generator can produce both alternating current and direct current.

- AC generator consists of slip rings, armature, magnet and brushes while DC generator consists of magnet commutator, armature and brushes.

► **Note** Electric motor is a rotating device which converts electrical energy into mechanical energy.

Transformer

Transformer is a device which converts low voltage AC into high voltage AC and high voltage AC into low voltage AC. It is based on mutual induction.

- Core of a transformer is made up of soft iron.
 - Step-up transformer converts a low voltage of high current into a high voltage of low current.
 - Step-down transformer converts a high voltage of low current into a low voltage of high current.
 - The main energy losses in a transformer are given below:
 - (a) Iron loss (b) Flux loss
 - (c) Hysteresis loss (d) Humming loss
 - Transformer is used in voltage regulators, refrigerators, computer, air conditioner.
 - Step-down transformer is used for welding purpose.
 - Audio transformer allowed telephone circuits to carry on a two-way conversation over a single pair of wires.
 - Radio frequency transformers are used in radio communication.
- **Note** A mobile phone charger is one of the best example of step down transformer as it extracts the power from the home supply (AC 220V) and convert it to DC of required voltage.

MODERN PHYSICS

CATHODE RAYS

If the gas pressure in a discharge tube is 10^{-2} to 10^{-3} mm of Hg and a potential difference of 10^4 V is applied between the electrode, then a beam of electrons emerges from the cathode which is called cathode rays.

- Cathode rays are invisible and travel in a straight line.
- These rays are deflected by electric and magnetic fields.
- These rays can ionise gases.
- They can produce chemical change and thus affect a photographic plate.
- When they strike a target of heavy metal such as tungsten, they produce X-rays.

X-RAYS

When cathode rays strike on a heavy metal of high melting points, then a very small fraction of its energy converts into a new type of waves, called X-rays.

- X-rays are electromagnetic waves and these produce photoelectric effect.

- X-rays were discovered by scientist Roentgen.
 - These produce illumination on falling the fluorescent substances and travel in a straight line with the speed of light.
 - X-rays penetrate through different depth into different substances.
 - Soft X-rays have greater wavelength and lower frequency and hard X-rays have lower wavelength and higher frequency.
 - X-rays ionise gases through which they pass.
 - X-rays are not deflected by electric and magnetic fields.
 - They show all the important properties of light rays like, reflection, refraction, interference, diffraction and polarisation, etc.
 - X-rays are used in surgery, radiotherapy, engineering department and searching.
- **Note** Barium and Iodine are some chemicals used as contrast mediums for X-ray based imaging method. i.e. they absorb X-ray to highlight the targeted body part.

POSITIVE OR CANAL RAYS

These rays are positively charged particles and are called positive rays or canal rays. These rays were discovered by Goldstein.

- These rays can affect the photographic plates.
- These rays are capable of producing physical and chemical changes.
- These rays are deflected by electric and magnetic fields. These rays can produce fluorescence and phosphorescence.

Fluorescence

Some substances absorb the light of higher frequency or shorter wavelength and emits a light having lower frequency or higher wavelength in the presence of light source. This phenomenon of emission of light is called **fluorescence**. Such substances are called fluorescence substances.

- Fluoresphor, petrol, uranium oxide and barium platino cyanide are examples of fluorescence substance.
- Barium platino cyanide is used in search of X-rays.

Compact Fluorescent Lamp (CFL)

Compact fluorescent lamp converts electrical energy into radiant energy.

- It has two components
 - A glass tube filled with argon and mercury vapour and coated with layer of fluorescent materials.
 - An electronic ballast circuit.
- The ballast circuit (small circuit board with rectifiers, a filter capacitor and usually two switching transistors) takes a 220V input from external power source and sends a current into the fluorescent tube as output.
- When power supply is given to the CFL, filament attached with the cathode heats up and emits electrons in the tube. This ionises the argon and mercury vapour particles, causing it, ultraviolet light emission.

Phosphorescence

Some substances after being kept in sunlight for sometime emit light even after the light source is removed. This phenomenon is called **phosphorescence**. Zinc sulphide, calcium sulphide and barium sulphide are examples of phosphorescence.

PHOTOELECTRIC EFFECT

When the ultraviolet rays (or the visible light of low wavelength or any other kind of electromagnetic radiation) are made incident on a metal, the electrons are emitted from it. This phenomenon is called the photoelectric emission or photoelectric effect.

Photoelectric cell is based on photoelectric effect. It is a device by which light energy is converted into the electrical energy.

Applications of photoelectric effect are given below

- (i) In reproduction of sound in cinema, television and photo telegraphy.
- (ii) In automatic switches for street lights.
- (iii) In photoelectric sorters.
- (iv) To control the temperature in furnace and in chemical process.
- (v) In industries for locating minor flows or holes in metallic sheets.

SEMICONDUCTOR

Semiconductors are those materials whose electrical conductivity at room temperature lies in between that of insulator and conductor. A semiconductor in an extremely pure form is known as intrinsic semiconductor.

- If a measured and small amount of chemical impurity is added to intrinsic semiconductor, it is called extrinsic semiconductor or doped semiconductor.
- External semiconductors are of two types
 - (i) *n*-type semiconductor
 - (ii) *p*-type semiconductor
- An extrinsic semiconductor in which electrons are majority charge carrier is called ***n*-type semiconductor**.
- An extrinsic semiconductor in which holes are the majority charge carrier is called ***p*-type semiconductor**.

Special Purpose Diodes

An arrangement consisting a *p*-type semiconductor brought into a close contact with *n*-type semiconductor is called a *p-n* junction or *p-n* junction diode.

- A device which converts alternating current (AC) or voltage into direct current (DC) or voltage is known as **rectifier**. DC can be converted into AC with the help of an inverter.
- LED's are specially designed diode made up of GaAsP, GaP and are used in electronic gadgets as indicator light. It is used as a rectifier which converts an alternating current into direct current.
- LED's light up very quickly. A typical red indicator LED will achieve full brightness in microseconds.
- LED or semiconductor laser is used to convert electrical energy into optical energy.
- Zener diode is a highly doped *p-n* junction diode which is not damaged by high reverse current. It can be used as voltage regulator.
- Photo diode is a special *p-n* junction diode fabricated with a transparent window to allow light to fall on the diode. It is used in reverse biasing.
- **Note** Diode valve acts as a rectifier and triode valve can be used as amplifier oscillator and detector.

Solar Cell

Solar cell is a *p-n* junction device which converts solar energy into electrical energy. It consists of a silicon or gallium arsenide *p-n* junction.

Uses of Solar Cell

- (i) Solar cells are used as solar cooker.
- (ii) Solar cells are used in wrist watches, calculators and light meters.
- (iii) Solar cells are also used in artificial satellite to provide power inside the satellite.
- (iv) Solar cells are used for charging storage batteries.

Integrated Circuits

An IC can be defined generally as an arrangement of multifunction semiconductor devices. It consists of a single crystal chip of silicon, nearly 1.5 mm^2 in cross-section.

TV remote control is an electronic device containing an IC and few other components. If a key on a remote control is pressed, it emits infrared signals which are received by the electronic circuit of the TV receiver, etc.

LASER

The word LASER is the short form of Light Amplification by Stimulated Emission of Radiation. It is an optical device, which produces an intensive beam highly coherent, monochromatic light.

Applications of laser rays are described below

- (i) Laser rays are used in treatment of detached retina, treatment of skin cancer and surgery of brain.
- (ii) Laser beams are used in signals and radio communications.
- (iii) The property of high intensity is used to cut hard metals.
- (iv) The weather informations are collected with the help of radar, an instrument based on the properties of laser.
- (v) The laser rays are parallel and do not spread. This property of laser rays is used in the measurement of long distance.

COMMUNICATION

The term communication refers to the transmitting receiving and processing of information by electronic means. A basic communication system consists of an information source, a transmitter, a link and a receiver.

- Electromagnetic waves which are used in radio, television and other communication system are radiowaves and micro waves.
- In **ground wave propagation**, the radio wave travel through atmosphere following the surface of the earth from transmitting antenna to receiving antenna. The ground wave propagation is useful for low range

frequencies upto 1500 kHz. The ground wave propagation is generally used for local broadcasting.

- In **space wave propagation**, the radio wave from the transmitting antenna reach the receiving antenna either directly or after reflection from the ground or from troposphere. The space wave propagation utilises the electromagnetic waves of very high frequency band (between 30 MHz to 300 MHz), ultra high frequency band (between 300 MHz to 3000 MHz) and microwaves having frequencies range 100 GHz to 300 GHz.
- In **sky wave propagation** the radio waves from the transmitting antenna reach the receiving antenna after entering in ionosphere. The sky wave frequency lies between 2 MHz to 30 MHz.
- A communication satellite is a space craft, provided with microwave receiver and transmitter.
 - It is placed in an orbit around the earth.
 - The frequencies used in satellite communication lies in UHF/ microwave regions.
 - These waves can cross the ionosphere and reach the satellite.
- The satellite communication is used effectively in mobile communication, metrology and weather forecasting.
- Remote sensing is the technique to obtain information about an object by observing it from a distance and without coming to actual contact with it. It is used in development of weather forecasting system.
- Parallel wire line is used to connect an antenna with TV receiver. Parallel wire lines are never used for transmission of microwaves.
- Twisted pair wires are used to connect telephone system.
- Coaxial cables are used for long distance high frequency transmission. e.g. TV channel transmission.
- **Note** Maser is a device working on the same principle as a laser. Maser means 'Microwave Amplification by Stimulated Emission of Radiation'. It is used to produce very high magnetic field and in radio astronomy and communications.

Optical Communication

It is a better communication system for large number of channels and higher bandwidth transmission. Now-a-days optical fibres of very small diameters ($\approx 10^{-5} \text{ m}$) are used in optical communication. Advantages of optical communication over the conventional two wires or coaxial cable systems are

- (i) In optical fibres energy losses are less.
- (ii) Secret information like banking, defence, etc., is more secure because optical signal is confined to the inside of fibre and cannot be tampered easily.
- (iii) It can be used at high frequencies ($\approx 10^{14} \text{ Hz}$).

Phenomenon of total internal reflection is the principle behind optical fibres. Light can be transmitted along it with almost no loss because of total internal reflection. Important applications of optical fibres are in telecommunications and medicine.